



Artificial Intelligence Foundations (CIS4049-N)

Week 2 - Lab Introduction

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Lab Session – Introduction

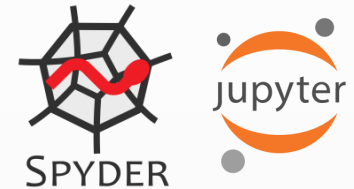
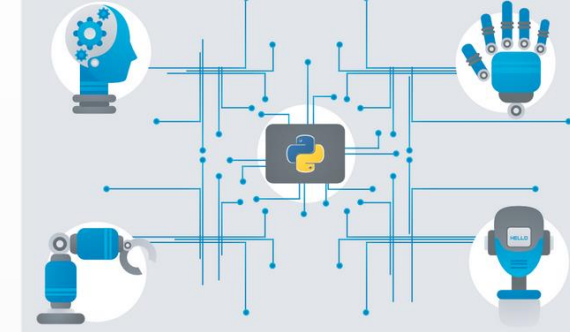
- **Introduction to Python and Python Basics**

- Introduction to Python and Installation Guide

Setting up the working environment

- Python Basics

- **Implementing Agents and Environments**



Why Python?

- One of the most **popular languages** (if not the most!) for **AI** and **ML**

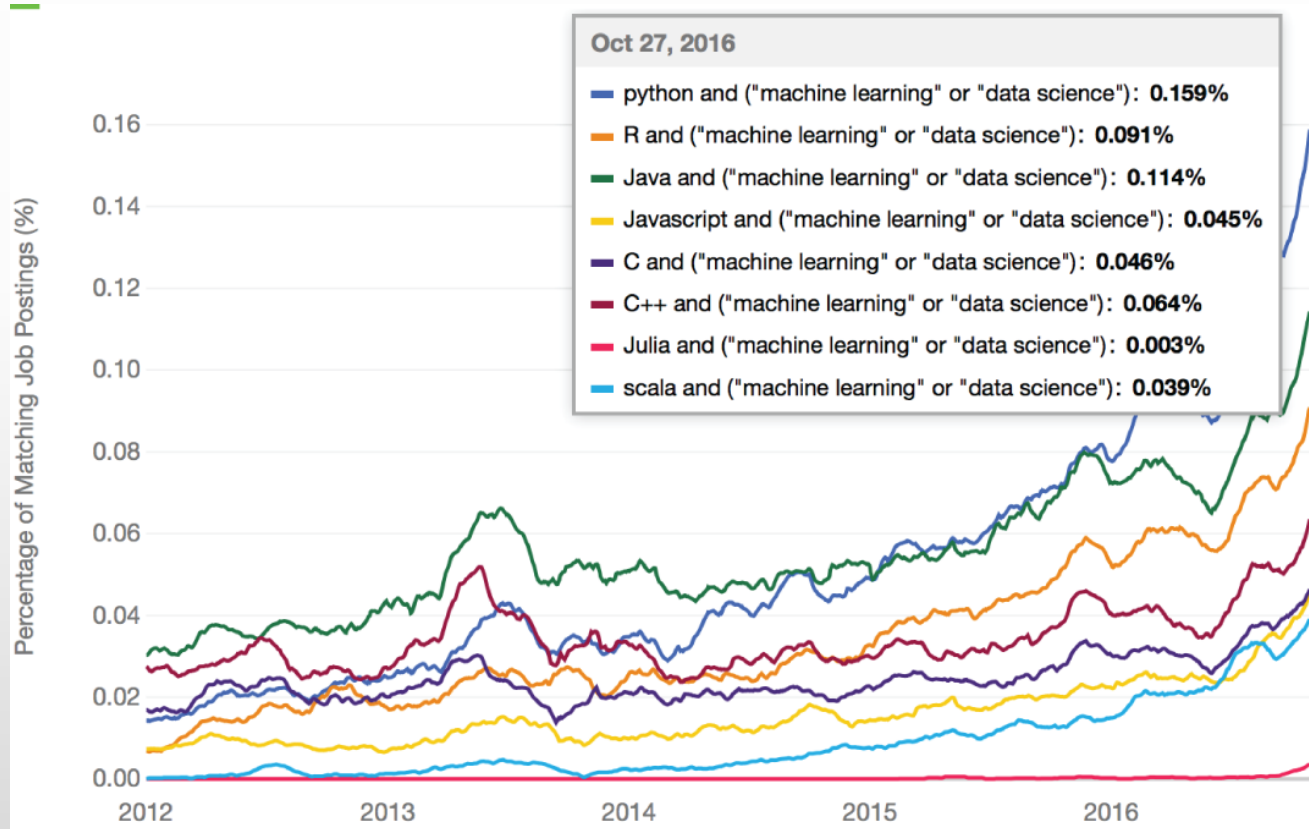


Image source: indeed.com

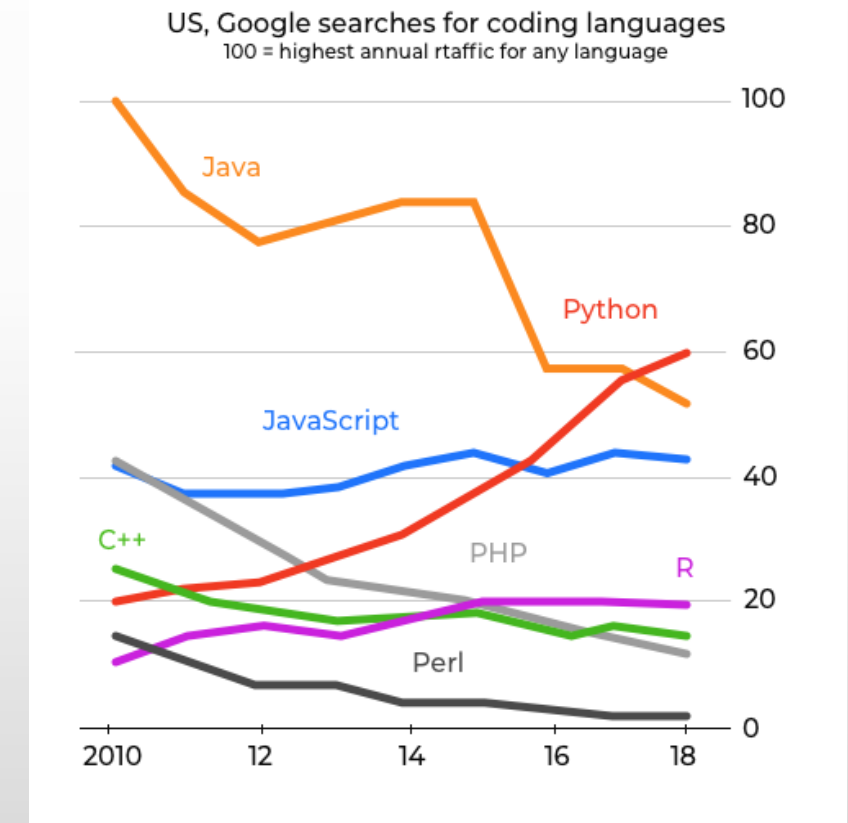


Image source: economist

Why Python?

- A great **library ecosystem**:
 - **Keras, TensorFlow, and Scikit-learn** for machine learning
 - **Pandas** for general-purpose data analysis
 - **NumPy** for scientific computing and data analysis
 - **SciPy** for advanced computing
 - **Matplotlib** and **Seaborn** for high-quality visualisations (line charts, histograms, scatter plots, and so on)
- **Simple to understand** and **consistent**: high-level programming language, easily readable syntax
- **Extremely efficient**: complete more work using fewer lines of code

Why Python?

- **Flexible:** the most flexible language in ML, reducing the possibility of errors
- **Platform-independent:** supported by many platforms including Linux, Windows, and macOS
- **Great community and Corporate Support:**
 - **Active and supportive community** (Python forums, etc.)
 - **Corporate support:** many top companies (Google, Facebook, Instagram, Netflix, Quora, etc.) use Python for their products. Google is single-handedly responsible for creating many of the Python libraries for Machine Learning such as Keras, TensorFlow, etc.
 - **Python AI community** has grown across the globe



Python

- Download Python 3 (currently using **Python 3.9.7**, released August 2021) from the official Python homepage: <https://www.python.org/downloads/>
- Install Python on your machine (Youtube Video Tutorial: <https://www.youtube.com/watch?v=UvcQIPZ8ecA>)
- Check if you have Python already installed on your machine (which is likely if you have a Macbook or a Linux machine) by typing the following command in a command shell:
python3 -v
- If you are working on Windows 10 (as in my case), type: **python3 -version**
- The output (in my case) is **Python 3.9.7**
- Now we are ready to set up the working environment with **Anaconda**

- **Anaconda** (<https://www.anaconda.com/>) is one of the mostly used **Python distributions**
- Anaconda provides the **Python interpreter**, together with a **list of Python packages** and sometimes other related tools, such as **editors** (*Spyder*)
- **Spyder** is an interactive **development environment** for Python, including an **editor**
- Download Anaconda from the official Anaconda homepage:
<https://www.anaconda.com/products/individual>
- Install Anaconda on your machine (Youtube Video Tutorial:
<https://www.youtube.com/watch?v=g6ln1dAt-RI&t=29s>)



Spyder



- **Spyder** - "Scientific **PY**thon **D**evelopment **E**nvi**R**onment" (<https://www.spyder-ide.org/>)
- Powerful **interactive development environment** for the Python language with advanced editing, interactive testing, debugging and introspection features
- We will use it as the **main environment** to learn about Python and ML
- Useful features include:
 - provision of the **IPython (Qt) console** as an interactive prompt, which can display plots inline
 - ability to **execute snippets of code** from the editor in the console
 - continuous **parsing of files** in editor, and provision of **visual warnings** about potential errors
 - **step-by-step execution**
 - **variable explorer**



Spyder



Spyder (Python 3.8)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\diste\OneDrive\Desktop

C:\Users\diste\OneDrive\Desktop\test-python-installation-2021.py

test-python-installation-2021.py x weather.py x dataframe_people.py x students_ML.py x housing.py x

```
1 # Python program that can be executed to report whether particular
2 # python packages are available on the system.
3
4 import math
5 import os
6 import sys
7
8
9 def test_is_python_35():
10     major = sys.version_info.major
11     minor = sys.version_info.minor
12     if major == 3:
13         pass
14     else:
15         print("You are running Python {}, but we need Python {}".format(major, 3))
16         print("Download and install the Anaconda distribution for Python 3.")
17         print("Stopping here.")
18
19     # Let's stop here
20     sys.exit(1)
21     return None
22     # assert major == 3, "Stopping here - we need Python 3."
23
24 if minor >= 5:
25     print("Testing Python version-> py{}.{}.format(major, minor))
26 else:
27     print("Warning: You should be running Python 3.5 or newer, " +
28           "you have Python {}.{}.format(major, minor))
29
30 def test_numpy():
31     try:
32         import numpy as np
33     except ImportError:
34         print("Could not import numpy -> numpy failed")
35         return None
36     # Simple test
37     a = np.arange(0, 100, 1)
38     assert np.sum(a) == sum(a)
39     print("Testing numpy... -> numpy OK")
40
41
42
```

1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00

sin(x)

x

Variable explorer Help Plots Files

Console 1/A x

```
diste/OneDrive/Desktop')
Running using Python 3.8.5 (default, Sep 3 2020, 21:29:08) [MSC v.1916 64 bit (AMD64)]
Testing Python version-> py3.8 OK
Testing numpy... -> numpy OK
Testing scipy... -> scipy OK
Testing pandas... -> pandas OK
Testing matplotlib... -> pylab OK
Testing sympy... -> sympy OK
Testing pytest... -> pytest OK

In [109]: |
```

IPython console History



Jupyter Notebook

- **“Jupyter”** - flask-based Python application that enables you to execute Python code in a browser. Instead of Python scripts, you can use Jupyter “notebooks,” which support various interactive features for executing Python code" (<https://jupyter.org/>)
- Jupyter’s name is a reference to the three core programming languages supported by Jupyter, which are **Julia**, **Python** and **R**, and also a homage to **Galileo**'s notebooks recording the discovery of the moons of Jupiter
- Already included in Anaconda
- You will start / stop the Jupyter Notebook server from the command line: **jupyter notebook**, and to stop the Jupyter Notebook server, just type: **Ctrl+C**
- **Google Colaboratory (colab)** supports Jupyter notebooks in a browser
- Useful video to understand how to work with Jupyter:
<https://www.youtube.com/watch?v=jZ952vChhul&t=112s>



https://en.wikipedia.org/wiki/Moons_of_Jupiter

Introduction to Python Basics

- **Introduction to Python**
- **Python basics** with code samples, and how to work with simple **data types**, such as numbers, fractions, and strings
- **Conditional logic, loops and functions:** how to work with conditional statements, loops, and user-defined functions in Python
- **Python Collections:** Lists (or arrays), sets, tuples, and dictionaries
- Many **short code blocks** that will help you rapidly **learn** or **review** how to work with these data structures in Python

Implementing Agents and Environments

- **Implementing Agents and Environments**
- **Agents.py** - <https://github.com/aimacode/aima-python/blob/master/agents.py>
- An **agent** is anything that can perceive its environment through sensors, and act upon that environment through actuators based on its agent program. This can be a dog, robot, or even you. As long as you can perceive the environment and act on it, you are an agent.
- **Python notebook on a blind dog** - How to implement a simple agent, create an environment, and create a program that helps the agent act on the environment based on its percepts.

<https://colab.research.google.com/github/tofighi/aima/blob/master/agents.ipynb#scrollTo=O23GWZAph8pk> -

- Importing all the functions from the agents.py module and create our first agent - a **blind dog**.

Wrap-up

- Python **installation** and set up the **working environment** (Anaconda, Spyder, Jupyter)
- Introduction to **Python for ML**: language basics (Week 2 - Lab session), **conditional logic**, **loops** and **functions**: how to work with conditional statements, loops, and user-defined functions in Python
- **Python Collections: Lists** (or arrays), sets, tuples, and dictionaries.
- **Implementing Agents and Environments** (e.g., blind dog)
- **General approach**: many short code blocks (scripts) that will help you rapidly learn or review how to work with these data structures in Python, and example scripts for implementing and customizing your AI solutions

Questions?

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